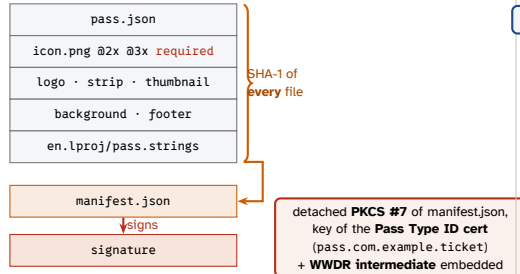
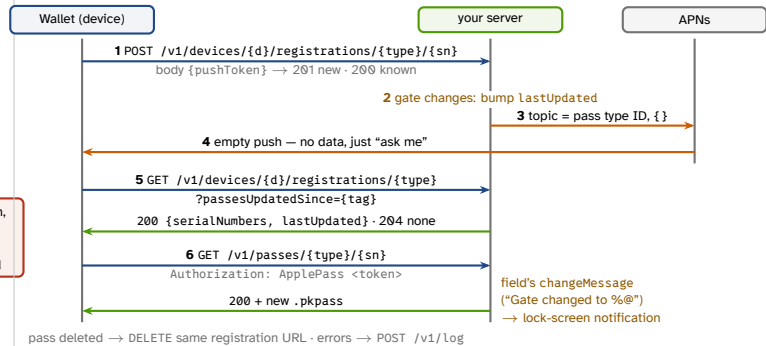


**In one line:** A Wallet pass is a **signed zip** (.pkpass): pass.json + images + manifest.json (SHA-1 of every file) + a **PKCS #7 signature** of the manifest made with your **Pass Type ID certificate**. Wallet surfaces it when **relevant** (time, place) and keeps it current via **your web service**: an **empty APNs push** says “changed”, the device *pulls*.

## Inside ticket.pkpass (a zip)



## Keeping a pass current — the web service



## How it works

- **One style key** in pass.json: boardingPass (+ transitType air/train/bus/boat/generic) · eventTicket · coupon · storeCard · generic. Each holds headerFields, primaryFields, secondaryFields, auxiliaryFields, backFields (key, label, value, changeMessage).
- **Richer layouts** via preferredStyleSchemes + semantics: poster event ticket (iOS 18, “posterEventTicket”), semanticBoardingPass (iOS 26, live flight status), poster generic (iOS 27). Keep the legacy fields — older iOS renders those.
- **Identity** = passTypeIdentifier + serialNumber. **Barcodes**: first one the device can show wins — QR, PDF417, Aztec, Code128 (iOS 27 adds EAN-13, Code 39, Codabar, ITF).
- **Relevance**: relevantDates (start/end; single relevantDate deprecated) · locations ( $\leq 10$ ) + maxDistance · beacons — the system decides to surface it.
- **Lifecycle**: expirationDate, voided (used coupon), associatedStoreIdentifiers + appLaunchURL.

## Example — pass.json (event ticket)

```
{ "formatVersion": 1,
  "passTypeIdentifier": "pass.com.example.ticket",
  "serialNumber": "T-000123", "teamIdentifier": "ABCDE12345",
  "organizationName": "Example Events", "description": "Concert ticket",
  "webServiceURL": "https://passes.example.com/",
  "authenticationToken": "<random secret per pass>",
  "relevantDates": [{ "startDate": "2026-10-10T19:00+02:00",
    "endDate": "2026-10-10T23:00+02:00" }],
  "barcodes": [{ "format": "PKBarcodeFormatQR", "message": "T-000123",
    "messageEncoding": "iso-8859-1" }],
  "eventTicket": {
    "primaryFields": [{ "key": "event", "label": "EVENT", "value": "The Band" }],
    "secondaryFields": [{ "key": "gate", "label": "GATE", "value": "B",
      "changeMessage": "Gate changed to %@"}] }
```

## Barcode vs NFC

A barcode is a picture — a screenshot works anywhere. An **NFC pass** adds nfc {message ( $\leq 64$  bytes), encryptionPublicKey (P-256), requiresAuthentication: the terminal speaks Apple’s **Value Added Services** protocol and only the reader holding the private key can read it. Needs a **special NFC entitlement from Apple** — you cannot self-enable it.

## Signing — the build step (server side)

1. Portal: register pass.com.example.ticket, download its **Pass Type ID certificate**.

2. Write manifest.json = {"pass.json": "<sha1 hex>", ...} for every file, localized ones included.
3. signature = detached PKCS #7 (CMS) of manifest.json, cert + private key, with the **WWDR intermediate** embedded.
4. Zip, rename .pkpass, serve with its MIME type → Safari/Mail offer *Add to Apple Wallet*.

The same cert sends the **update pushes**: expired → no new passes *and* no updates.

## In the app — PassKit

```
let pass = try PKPass(data: pkpassData) // throws on a bad signature
let lib = PKPassLibrary()
if lib.containsPass(pass) { _ = lib.replacePass(with: pass) }
else if let vc = PKAddPassesViewController(pass: pass) {
  present(vc, animated: true) } // user confirms "Add"
```

addPasses(…) for several; PKAddPassButton for the badge. Reading passes (passes()) needs the com.apple.developer.pass-type-identifiers entitlement and returns **only your** pass types.

## Verify with Wallet (identity)

PKIdentityAuthorizationController (iOS 16, iPhone, special entitlement): ask for fields of an ID in Wallet; the user approves; an **encrypted CBOR** response goes to your **server** to decrypt.

## Traps

- ! Expecting data in the push: it is {}; forget to bump lastUpdated → nothing pulled.
- ! Editing a file after hashing, or one missing from the manifest → invalid pass.
- ! No WWDR intermediate in the signature → rejected with no detail.
- ! New serialNumber for the same ticket = a second pass, not an update.
- ! sharingProhibited hides Share, it is not DRM; a QR screenshots fine — NFC doesn’t.
- ! Poster ticket without legacy fields → blank pass on older iOS.

## Remember

**“Hash every file, sign the manifest; push nothing, pull everything.”**